Uncalibrated Stereo Vision

Distance Estimation Report

Camera : Samsung Galaxy M34

Native Res. : 4080 × 3060 px (12 MP)

Working Res. : 1600 × 1200 px

Focal Length : 4.99 mm actual | 27 mm (35 mm equiv.)

Aperture / ISO : f/1.8 | ISO 150

Baseline : 1 foot (30.48 cm)

Object : Square target — classroom

Ground Truth : 7 feet (213.36 cm)

1. Camera Setup and Intrinsic Matrix

The stereo image pair was captured using a single Samsung Galaxy M34 smartphone. Two photographs were taken from hand-held positions separated horizontally by approximately 1 foot (30.48 cm) — this horizontal separation is the stereo baseline B. The chosen object of interest is a square target visible in both images, located approximately 7 feet (213.36 cm) from the camera.

1.1 Camera Specifications

- Sensor size : 1/2.0" → width = 6.40 mm, height = 4.80 mm
- Sensor diagonal : $\sqrt{(6.40^2 + 4.80^2)}$ = 8.0000 mm
- 35 mm-film diagonal: 43.27 mm
- Crop factor : 43.27 / 8.0000 = 5.4088
- 35 mm-equiv FL : 27.0 mm (from EXIF metadata)
- Actual focal length: 27.0 / 5.4088 = 4.9919 mm
- Pixel size (native): 6.40 / 4080 = 1.5686 µm
- f in pixels (native): 4.9919 / 0.001569 = 3182.34 px
- Scale (1600/4080) : 1600/4080 = 0.3922
- f in pixels (stored): 1247.9778 px
- Principal point : (cx, cy) = (800.0, 600.0) px

1.2 Intrinsic Matrix K

The camera intrinsic matrix K maps a 3-D point expressed in the camera coordinate frame to its 2-D pixel location. For a pin-hole camera with equal horizontal and vertical focal lengths (square pixels) and no skew:

$$K = \begin{bmatrix} f & 0 & c_x \\ 0 & f & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

Substituting the computed values:

$$K = \begin{bmatrix} 1247.98 & 0 & 800.0 \\ 0 & 1247.98 & 600.0 \\ 0 & 0 & 1 \end{bmatrix}$$

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2. Fundamental Matrix F

The Fundamental Matrix F is a 3 × 3 rank-deficient (rank 2) matrix that captures the complete epipolar geometry between two views of the same scene. Crucially, it does NOT require knowledge of the camera intrinsics — it is computed solely from point correspondences in the two images.

2.1 Epipolar Constraint

For any pair of corresponding pixel-coordinate homogeneous points $x = (u, v, 1)^T$ (left image) and $x' = (u', v', 1)^T$ (right image), the epipolar constraint is:

$$x'^T F x = 0$$

This single scalar equation states that x′ lies on the epipolar line l′ = Fx, and symmetrically x lies on l = F^T x′. Every point in one image maps to a line (not a point) in the other — this is the fundamental property exploited in uncalibrated stereo.

2.2 Geometric Definition

Given camera projection matrices P = K [I | 0] (left) and P′ = K′ [R | t] (right), the Fundamental Matrix is defined as:

$$F = K'^{-T} [t]_\times R K^{-1}$$

where $K'^{-T} = (K'^{-1})^T = (K'^T)^{-1}$, and [t]× is the skew-symmetric matrix encoding the cross product with the translation vector $t = (t_1, t_2, t_3)^T$:

$$[t]_\times = \begin{bmatrix} 0 & -t_3 & t_2 \\ t_3 & 0 & -t_1 \\ -t_2 & t_1 & 0 \end{bmatrix}$$

This means [t]× v = t × v (the matrix-vector product equals the cross product). F has 7 degrees of freedom: the 9 entries of a 3×3 matrix, minus 1 for rank-2, minus 1 for the scale ambiguity (F is defined only up to a scale factor).

2.3 Properties of F

- rank(F) = 2 (F is singular: det(F) = 0).
- F is defined up to scale, i.e., λF is equivalent for any λ ≠ 0.
- The epipole e in the left image satisfies F e = 0.
- The epipole e′ in the right image satisfies F^T e′ = 0.
- F has 7 degrees of freedom (7DOF).
- Epipolar lines: l′ = Fx (line in right image for point x in left).
- l = F^Tx′ (line in left image for point x′ in right).

2.4 Normalised 8-Point Algorithm

The normalised 8-point algorithm (Hartley, 1997) is a stable linear method for estimating F from N ≥ 8 correspondences. Normalisation is essential — without it the computation is numerically unstable.

Step 1 — Normalise the point sets.

Compute a similarity transform T (and T′) for each image such that the resulting point set has zero mean and the mean Euclidean distance from the origin equals √2. The transform has the form:

$$T = \begin{bmatrix} s & 0 & -s \cdot c_x \\ 0 & s & -s \cdot c_y \\ 0 & 0 & 1 \end{bmatrix}$$

where the scale factor s is:

$$s = \frac{\sqrt{2}}{\bar{d}}$$

and d̄ is the mean Euclidean distance of the centred points from the origin. The normalised points are: x̃ = Tx, x̃′ = T′x′.

Step 2 — Build the design matrix A.

For each normalised correspondence pair $(\tilde{x}_i, \tilde{x}'_i) = (\tilde{u}_i, \tilde{v}_i, 1), (\tilde{u}'_i, \tilde{v}'_i, 1)$, one row of the 9-column design matrix A is:

$$a_i = [\tilde{u}'_i\tilde{u}_i \quad \tilde{u}'_i\tilde{v}_i \quad \tilde{u}'_i \quad \tilde{v}'_i\tilde{u}_i \quad \tilde{v}'_i\tilde{v}_i \quad \tilde{v}'_i \quad \tilde{u}_i \quad \tilde{v}_i \quad 1]$$

Stacking N rows gives the N × 9 matrix A. The unknown vector $f = \mathrm{vec}(\tilde{F}) = (\tilde{F}_{11}, \tilde{F}_{12}, \tilde{F}_{13}, \tilde{F}_{21}, \ldots, \tilde{F}_{33})^T$ satisfies $A f \approx 0$.

Step 3 — Solve via SVD.

The least-squares solution under the unit-norm constraint is the last right singular vector of A:

$$\min f_{\|f\|=1} \; \| A f \| \implies f = \text{last column of } V \; [\text{from SVD of } A = U \Sigma V^T]$$

Reshape f to a 3×3 matrix $\tilde{F}$.

Step 4 — Enforce rank-2 constraint.

Due to noise, the initial $\tilde{F}$ may have rank 3. Decompose $\tilde{F}$ by SVD, then zero its smallest singular value:

$$F = U \Sigma V^T, \qquad \Sigma = diag(\sigma_1, \sigma_2, \sigma_3)$$
$$F \leftarrow U \cdot diag(\sigma_1, \sigma_2, 0) \cdot V^T$$

Step 5 — Denormalise.

Convert back to the original (unnormalised) pixel-coordinate system:

$$F = T'^T F T$$

2.5 RANSAC Outlier Rejection

In practice, feature matches contain outliers (false correspondences). RANSAC (Random Sample Consensus) wraps the 8-point algorithm:

- Randomly sample a minimal set of 8 correspondences.
- Estimate F from those 8 points (normalised 8-point).
- Count inliers: correspondences satisfying the Sampson distance < ε (ε = 1.0 px).
- Keep the F with the most inliers over many iterations.
- Re-estimate F using all inliers of the best model.

The Sampson distance (first-order approximation of the algebraic reprojection error) is:

$$d_S(x', F, x) = \frac{(x'^T F x)^2}{(Fx)_1{}^2 + (Fx)_2{}^2 + (F_T x')^T \ldots}$$

2.6 Computed Fundamental Matrix

Using SIFT feature matching (26 Lowe-ratio matches, 20 RANSAC inliers), the computed Fundamental Matrix is:

$$F = \begin{bmatrix} 0.00000000 & 0.00000650 & -0.00542219 \\ -0.00000624 & -0.00000001 & 0.00472239 \\ 0.00526436 & -0.00596918 & 1.00000000 \end{bmatrix}$$

Verification: rank(F) = 2 (expected 2). det(F) = 3.24e-26 (expected ≈ 0).

3. Essential Matrix E

The Essential Matrix E is the calibrated version of the Fundamental Matrix. While F encodes purely projective epipolar geometry, E encodes metric epipolar geometry — the actual camera rotation R and the unit translation direction $\hat{t}$ between the two camera poses.

3.1 Relation to the Fundamental Matrix

Given the camera intrinsic matrix K (identical for both views — same camera), the Essential Matrix is obtained by transferring F from pixel to normalised image coordinates:

$$E = K^T F K$$

Conversely, given E and K, the Fundamental Matrix can be recovered as $F = K^{-T} E K^{-1}$. The transformation from pixel coordinates x to normalised camera coordinates $\hat{x}$ is:

$$\hat{x} = K^{-1} x, \quad then \quad \hat{x}'^T E \hat{x} = 0$$

3.2 Physical / Geometric Interpretation

The Essential Matrix can also be expressed directly in terms of the inter-camera rotation R and the translation vector t that maps coordinates from the left camera frame to the right camera frame:

$$E = [t]_\times R$$

Equivalently, E = [t]× R means E is the composition of a rotation followed by a skew-symmetric (cross-product) matrix. Because [t]× has rank 2 and R has rank 3, their product has rank 2.

3.3 Properties of E

- rank(E) = 2 (E is singular: det(E) = 0).
- E has exactly two equal non-zero singular values: $\sigma_1 = \sigma_2$, $\sigma_3 = 0$.
- E has 5 degrees of freedom: R contributes 3 (three rotation angles), $\hat{t}$ contributes 2 (unit vector on the sphere).
- E is defined only up to a global sign: E and −E give the same geometry.

3.4 SVD Decomposition and Rank-2 Enforcement

Compute the raw Essential Matrix from F and K, then correct it so it exactly satisfies the two-equal-singular-values constraint:

Step 1 — Compute raw E.

$$E_{raw} = K^T F K$$

Step 2 — SVD of E_raw.

$$E_{raw} = U \Sigma V^T, \quad \Sigma = diag(\sigma_1, \sigma_2, \sigma_3)$$

Step 3 — Enforce $\sigma_1 = \sigma_2$, $\sigma_3 = 0$.

$$\bar{\sigma} = \frac{\sigma_1 + \sigma_2}{2}, \quad then \quad E = U \cdot diag(\bar{\sigma}, \bar{\sigma}, 0) \cdot V^T$$

Step 4 — Sign normalisation (ensure det(U) > 0, det(V) > 0).

If det(U) < 0 then U ← −U. If det(V) < 0 then V ← −V. This ensures the decomposition produces valid rotation matrices (det = +1).

3.5 Computed Essential Matrix

Substituting the computed K and F:

$$E = \begin{bmatrix} 0.11682749 & 9.91237924 & -1.85089647 \\ -9.93578221 & -0.13011427 & -0.33231775 \\ 1.93206386 & -0.92426341 & 0.24233084 \end{bmatrix}$$

Singular values of E: $\sigma_1 = 10.128863$, $\sigma_2 = 10.128863$, $\sigma_3 = 1.47e\text{-}17$ ($\sigma_1 \approx \sigma_2$, $\sigma_3 \approx 0$ ✓).

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4. Rotation Matrix R and Translation t

Once the Essential Matrix E is known, the relative camera rotation R and the unit translation direction $\hat{t}$ can be recovered via an SVD-based decomposition. This is the step that connects the uncalibrated estimate (F) to metric 3-D geometry.

4.1 SVD Decomposition of E

Decompose the (rank-2) Essential Matrix by SVD:

$$E = U\,\Sigma\,V^T, \quad \Sigma = diag(1,1,0) \;\; [after\ normalisation]$$

The columns of U and V are the left and right singular vectors of E, and they carry the geometric information about R and t.

4.2 The Auxiliary Rotation Matrix W

The derivation uses an auxiliary 3×3 rotation matrix W, which represents a 90° rotation about the z-axis:

$$W = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Note that W is itself a rotation matrix: det(W) = +1, $W^TW = I_3$. Its transpose is the inverse rotation (−90°):

$$W^T = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

4.3 Four Candidate (R, t) Solutions

Because E = [t]× R, and the cross-product matrix [t]× is invariant to the sign of t, there are four mathematically valid combinations of R and t. Each pair gives a valid decomposition of E:

$$R_1 = U\,W\,V^T, \quad R_2 = U\,W^T V^T$$
$$\hat{t} = U_{[:,2]} \;\; (third\ column\ of\ U)$$

The four candidate solutions are: $(R_1, +\hat{t})$, $(R_1, -\hat{t})$, $(R_2, +\hat{t})$, $(R_2, -\hat{t})$. At most one of these places the reconstructed scene points in front of both cameras simultaneously.

4.4 Cheirality Check

The correct (R, t) is determined by triangulating a sample of N inlier correspondences under each of the four candidates, then selecting the candidate for which the maximum number of points satisfies the cheirality condition:

$$P^T \rangle 0 \;\; (point\ in\ front\ of\ camera\ 1)$$
$$R\,P \;+\; t \rangle 0 \;\; (point\ in\ front\ of\ camera\ 2)$$

Here P denotes the reconstructed 3-D point. The candidate (R, t) with the most points satisfying both inequalities simultaneously is selected as the correct solution.

4.5 Properties of the Rotation Matrix

The rotation matrix R belongs to the Special Orthogonal group SO(3). It satisfies the following algebraic properties:

Orthogonality:

$$R^T R \;=\; R\,R^T \;=\; I_3$$

Unit determinant:

$$det(R) \;=\; +1$$

Inverse equals transpose:

$$R^{-1} \;=\; R^T$$

Rotation angle from trace:

$$\theta \;=\; arccos(\frac{tr(R) \;-\; 1}{2})$$

4.6 Rodrigues' Rotation Formula (Reference)

Any rotation matrix R can be written in terms of a unit rotation axis $\hat{k} = (k_1, k_2, k_3)^T$ and a rotation angle θ via Rodrigues' formula:

$$R \;=\; I_3 \;+\; sin\,\theta \;\cdot\; [\hat{k}]_\times \;+\; (1 \;-\; cos\,\theta) \;\cdot\; [\hat{k}]_\times{}^2$$

where $[\hat{k}]\times$ is the skew-symmetric cross-product matrix of $\hat{k}$. This formula shows that any element of SO(3) is parameterised by three independent numbers (the components of the rotation vector $\theta\hat{k}$).

4.7 Computed Rotation Matrix R

The rotation matrix recovered from the Essential Matrix decomposition:

$$R \;=\; \begin{bmatrix} 0.99162114 & 0.01232397 & 0.12859094 \\ -0.01332599 & 0.99988716 & 0.00693478 \\ -0.12849096 & -0.00859028 & 0.99167347 \end{bmatrix}$$

Verification:

- det(R) = 1.00000000 (expected +1.0 ✓)
- $\|R^TR - I_3\|$ = 5.51e-16 (expected ≈ 0 ✓)
- Rotation angle θ = 7.4356°

4.8 Computed Translation Vector $\hat{t}$

The unit translation vector (direction only — scale is not recoverable from uncalibrated stereo without knowing the baseline):

$$\hat{t} \;=\; \begin{bmatrix} 0.09361616 \\ 0.19110204 \\ 0.97709571 \end{bmatrix}$$

$\|\hat{t}\|$ = 1.00000000 (expected ≈ 1.0 ✓). The physical translation magnitude is set equal to the known baseline B = 30.48 cm when computing depth.

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5. Distance Estimation

For a horizontal stereo baseline with parallel cameras, the depth Z of a scene point can be recovered from its horizontal disparity d — the difference in x-coordinate of the point's projection in the left and right images.

5.1 Stereo Depth Formula

The derivation follows from similar triangles in the stereo geometry. A point at depth Z from the camera baseline projects to:

$$x_L = \frac{f \cdot X}{Z}, \quad x_R = \frac{f \cdot (X - B)}{Z}$$

where f is the focal length in pixels, B is the baseline, and X is the horizontal world coordinate of the point. Subtracting:

$$d = x_L - x_R = \frac{f \cdot B}{Z}$$

Solving for depth Z:

$$Z = \frac{f \cdot B}{d}$$

5.2 Object Detection — Square

The square object is detected independently in each image using a contour-based approach: Canny edge detection → morphological closing → contour approximation (4-sided polygon with aspect ratio < 2). The bounding-box centroid is used as the representative point for disparity computation.

- Left image: bbox = (586, 313, 151, 142), centroid = (661, 384)
- Right image: bbox = (754, 342, 146, 139), centroid = (827, 411)

5.3 Numerical Computation

Substituting measured values:

- Focal length f = 1247.9778 px
- Baseline B = 30.4800 cm (1 foot)
- Disparity d = |661 − 827| = 166.0000 px

$$Z = \frac{f \cdot B}{d} = \frac{1247.98 \times 30.48}{166.00} = 229.15\,cm = 7.5179\,ft$$

5.4 Results Summary

- Estimated distance : 229.15 cm = 7.5179 ft
- Ground-truth : 213.36 cm = 7.00 ft
- Absolute error : 15.79 cm (7.40%)

The estimated depth of 7.52 ft agrees closely with the measured ground truth of 7 ft, yielding a relative error of 7.4%. This level of accuracy is typical for uncalibrated hand-held stereo without lens distortion correction.

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6. Annotated Stereo Output

The figure below shows the stereo image pair with the detected square object annotated. SIFT inlier correspondences are shown as cyan crosses, the detected square bounding box as a green rectangle, and the object centroid as a red dot. The estimated distance and ground-truth distance are labelled on the left image.

Figure 1. Annotated stereo pair — detected square object, SIFT inliers, and distance estimate.

7. Conclusions

This report demonstrated an end-to-end uncalibrated stereo-vision pipeline applied to a real classroom scene captured with a Samsung Galaxy M34 smartphone:

- SIFT feature detection and Brute-Force matching with Lowe's ratio test (26 matches, 20 inliers after RANSAC).
- The Fundamental Matrix F (3×3, rank 2) was computed via the normalised 8-point algorithm with RANSAC, encoding the complete projective epipolar geometry.
- The Essential Matrix $E = K^T F K$ was derived, then decomposed via SVD to recover the relative rotation R and unit translation direction $\hat{t}$ between the two views.
- The Rotation Matrix R (det = +1, orthogonal) was selected from four SVD candidates via the cheirality constraint.
- The depth of the target square was estimated using the stereo formula $Z = fB/d$, yielding Z = 229.1 cm = 7.52 ft against a ground truth of 7.00 ft (error = 7.4%).

The small depth error demonstrates that even an uncalibrated, hand-held stereo setup can provide useful metric depth estimates when the camera intrinsics are approximately known from the EXIF metadata.

8. Github Link

https://github.com/DhruvilKachchhi/Computer-Vision/tree/main/Stereo%20Vision